



Human Computer Interaction

Introduction to the course

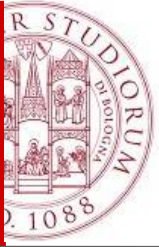
Prof. Catia Prandi
catia.prandi@unibo.it

Disclaimers



Disclaimers





Current position

- Associate Professor, Department of Computer Science and Engineering, at the University of Bologna
- Research Fellow at ITI (Interactive Technologies Institute)
 - Member of the Sense&Tell lab, Making Sense of Complex Data through Storytelling and Interactive Media
- My research area is: **Human-Computer Interaction (HCI)**

Subject Areas



My research

Keywords

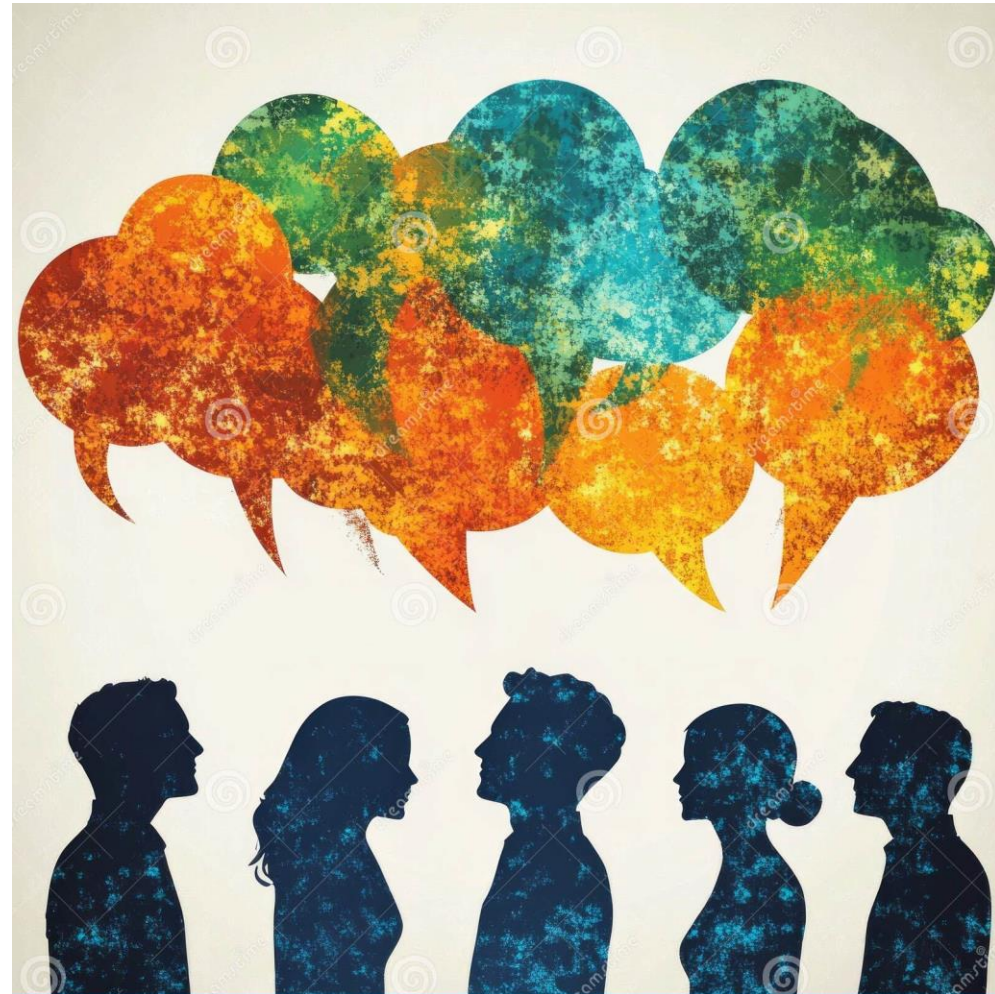


- My research through case studies
 - I. Urban accessibility and personalized paths
 - II. Sustainable mobility and tourism
 - III. Biodiversity awareness and monitoring
 - IV. University community and sustainability
 - v. Teenagers Well-being

Disclaimers

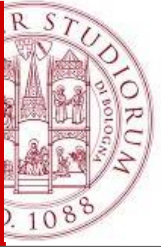


Disclaimers



Human Computer Interaction ?

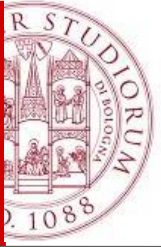




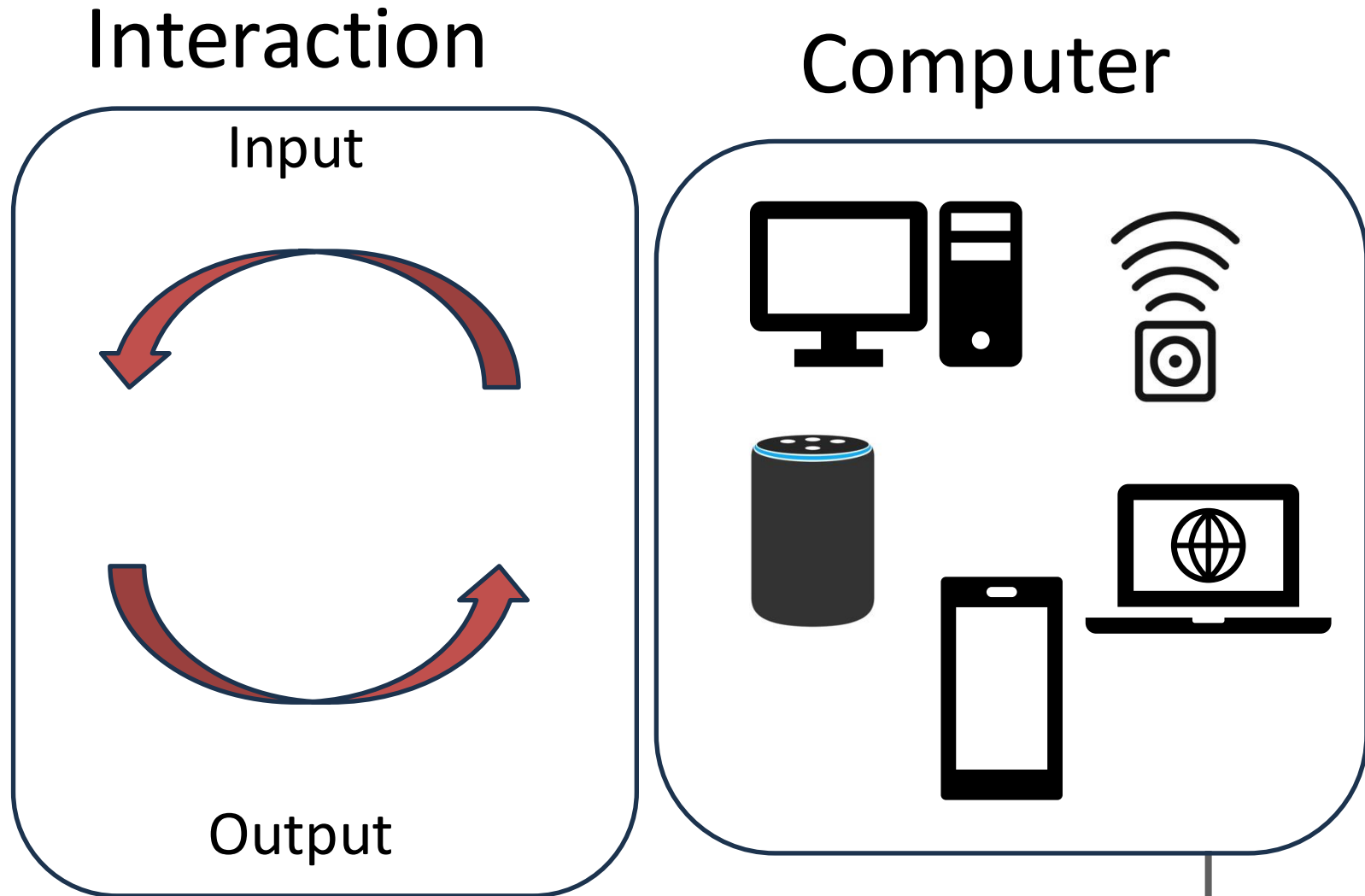
(No) Human-Computer Interaction

Computer





(No) Human-Computer Interaction



(No) Human-Computer Interaction

Human



Interaction

Input



Output



Computer



Human-Computer Interaction

Human



We talk first with the Human, then
decide the Technology after and then
the Interaction with the Technology

Human-Computer Interaction

Human



Computer



Human-Computer Interaction

Human



Interaction

Input

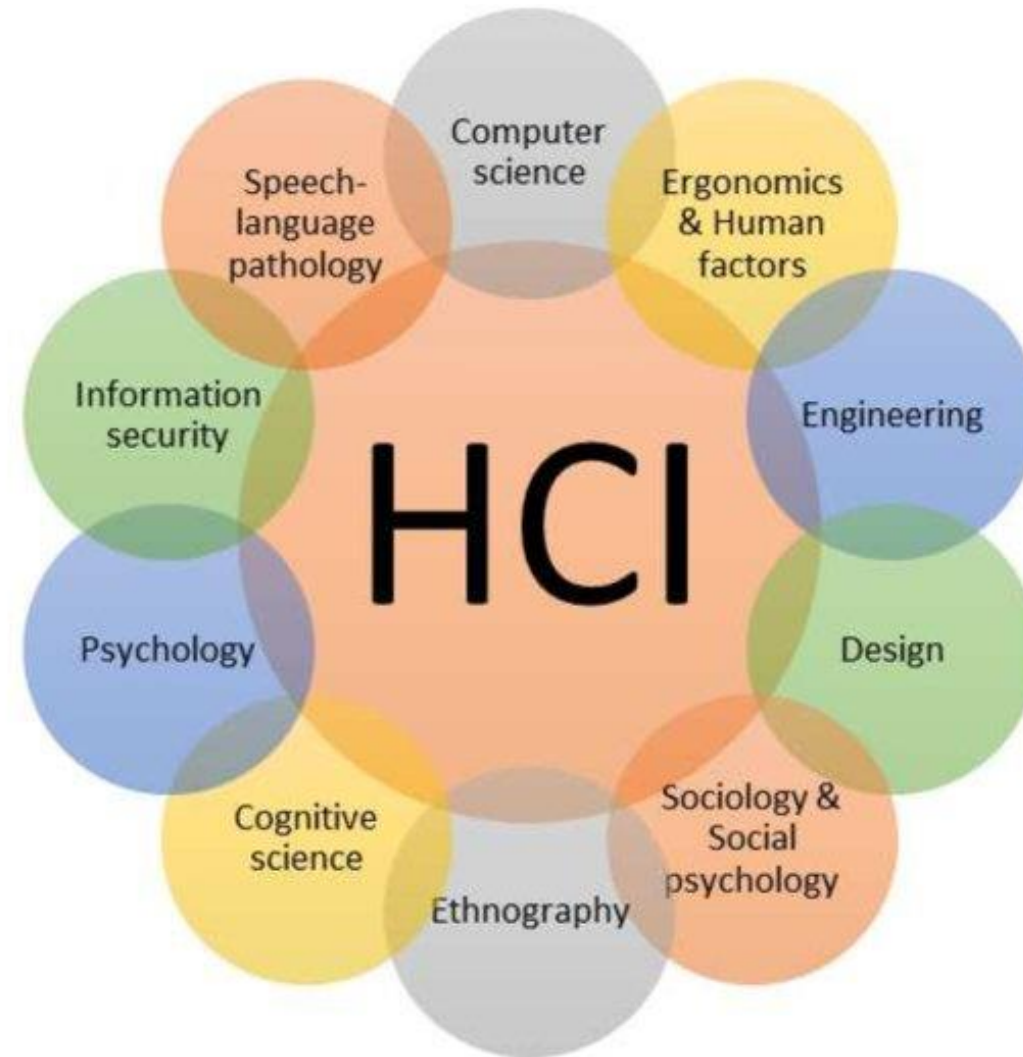


Output



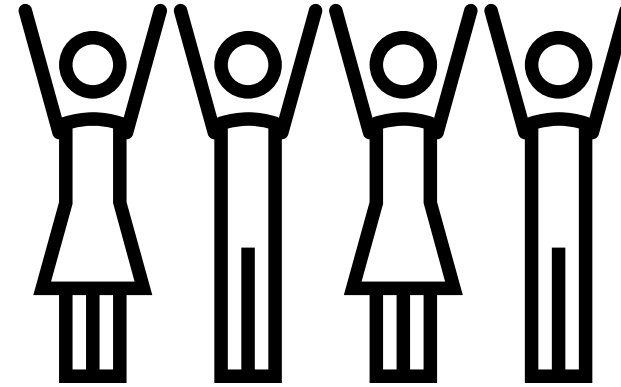
Computer

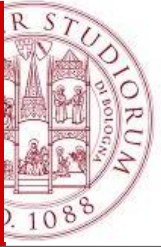




Agenda

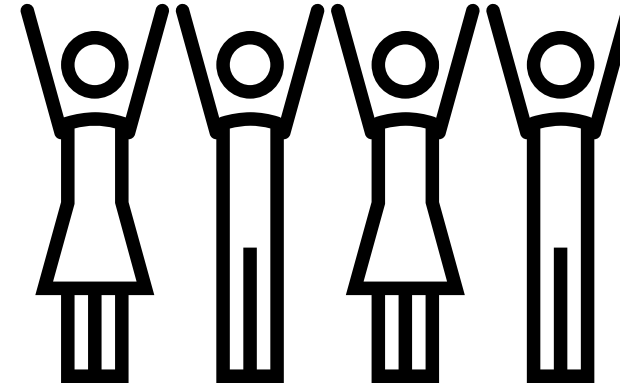
**Students
(you!)**
Expectations?





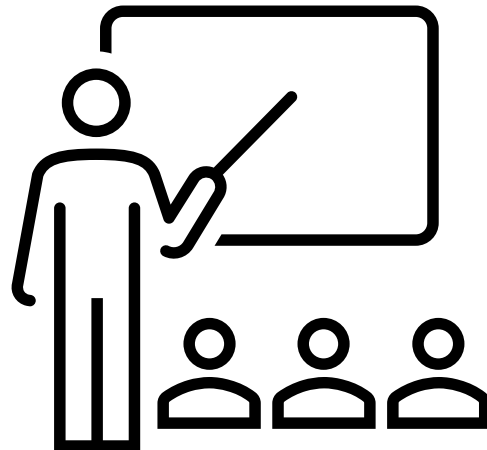
Agenda

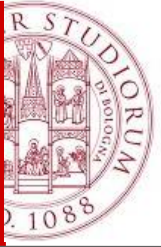
Students
(you!)
Expectations?



Teacher (me)

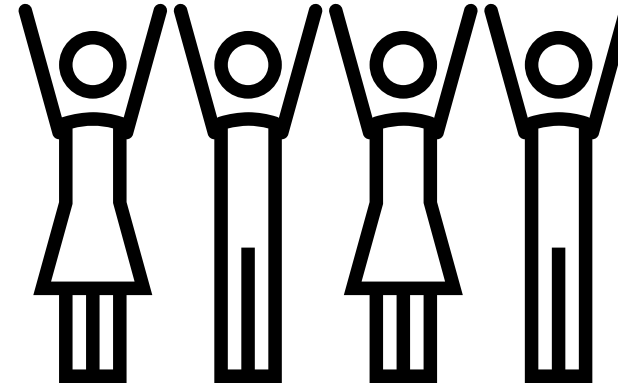
*Goals and
motivation*



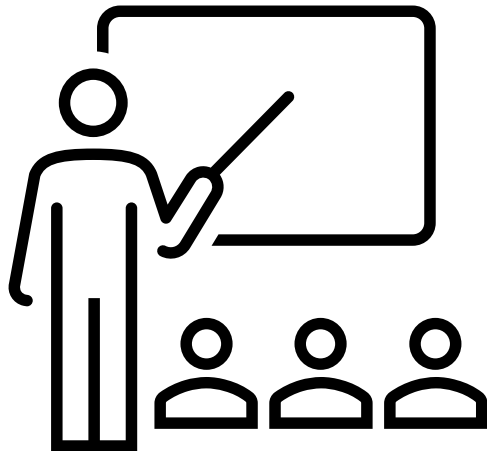


Agenda

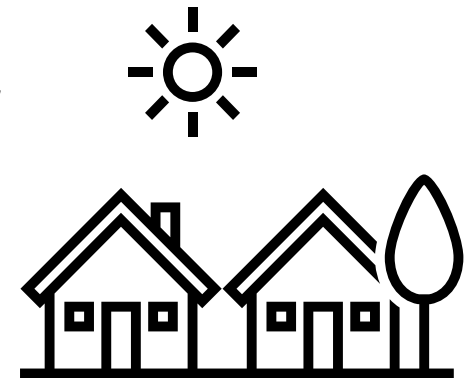
**Students
(you!)**
Expectations?



Teacher (me)
*Goals and
motivation*

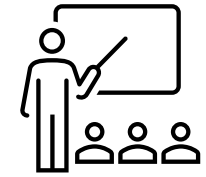


Everybody here
*Topics, organization, and
exam*





What do you hope to learn in this course?



Goals and Motivation



Goals

From user need to Low, medium and High fidelity prototype

1

Understanding how to **design the user experience** when interacting with modern applications, devices, and environments

2

Gaining in-depth knowledge of a **human-centered process** to create interactive systems and how to apply it in practice

3

Becoming familiar with methods to **gather and listen to users' needs**

4

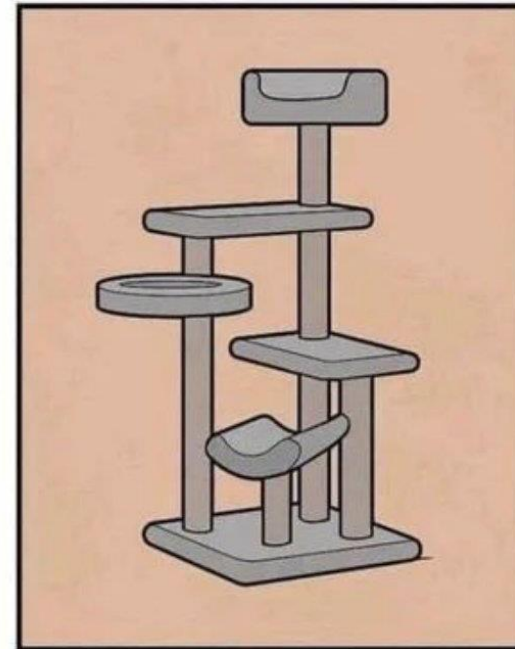
Learning to **evaluate** interactive systems with their users

Why?

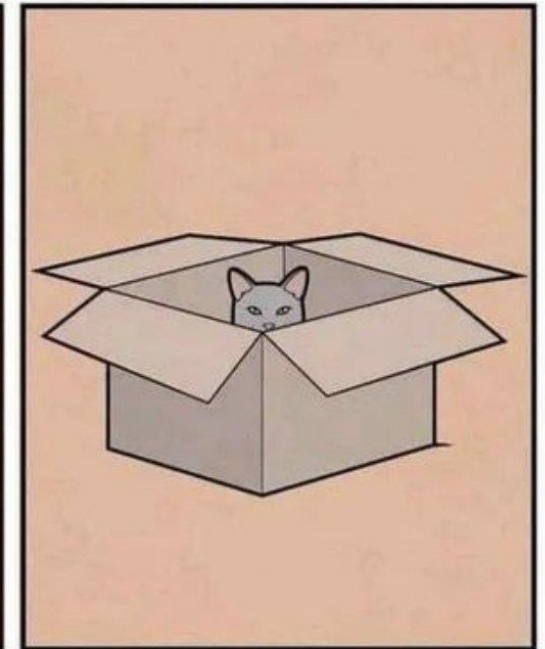


source: https://www.instagram.com/p/CT8qVYaDE_R/

Product features

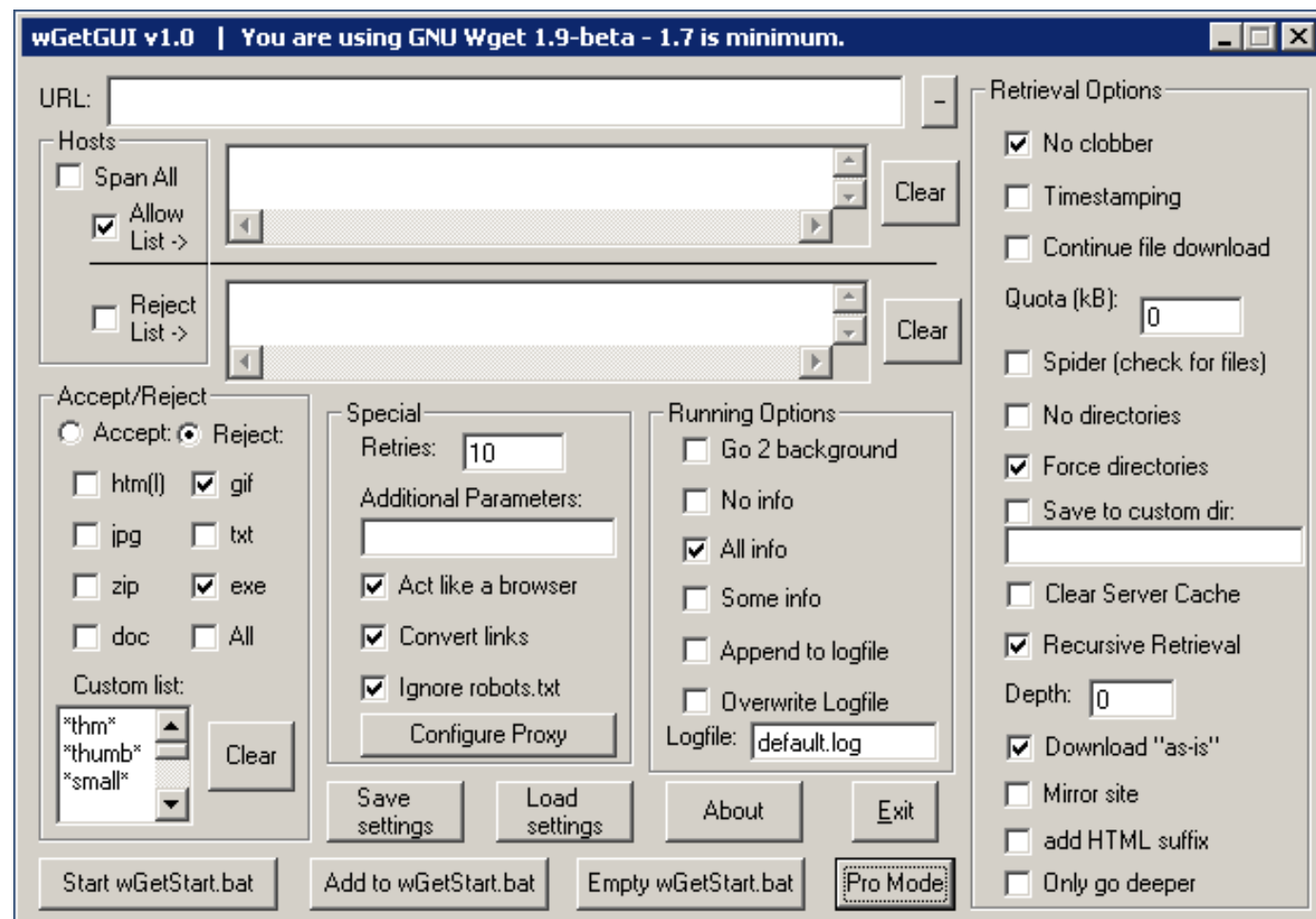


User needs



@_yes_but

Why?

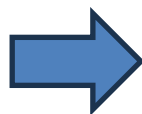




Why?

Bad example

GNU Wget is a free software package for retrieving files using HTTP, HTTPS, FTP and FTPS (the most widely used Internet protocols). It is a **non-interactive commandline tool**, so it may easily be called from scripts, cron jobs, terminals without X-Windows support.

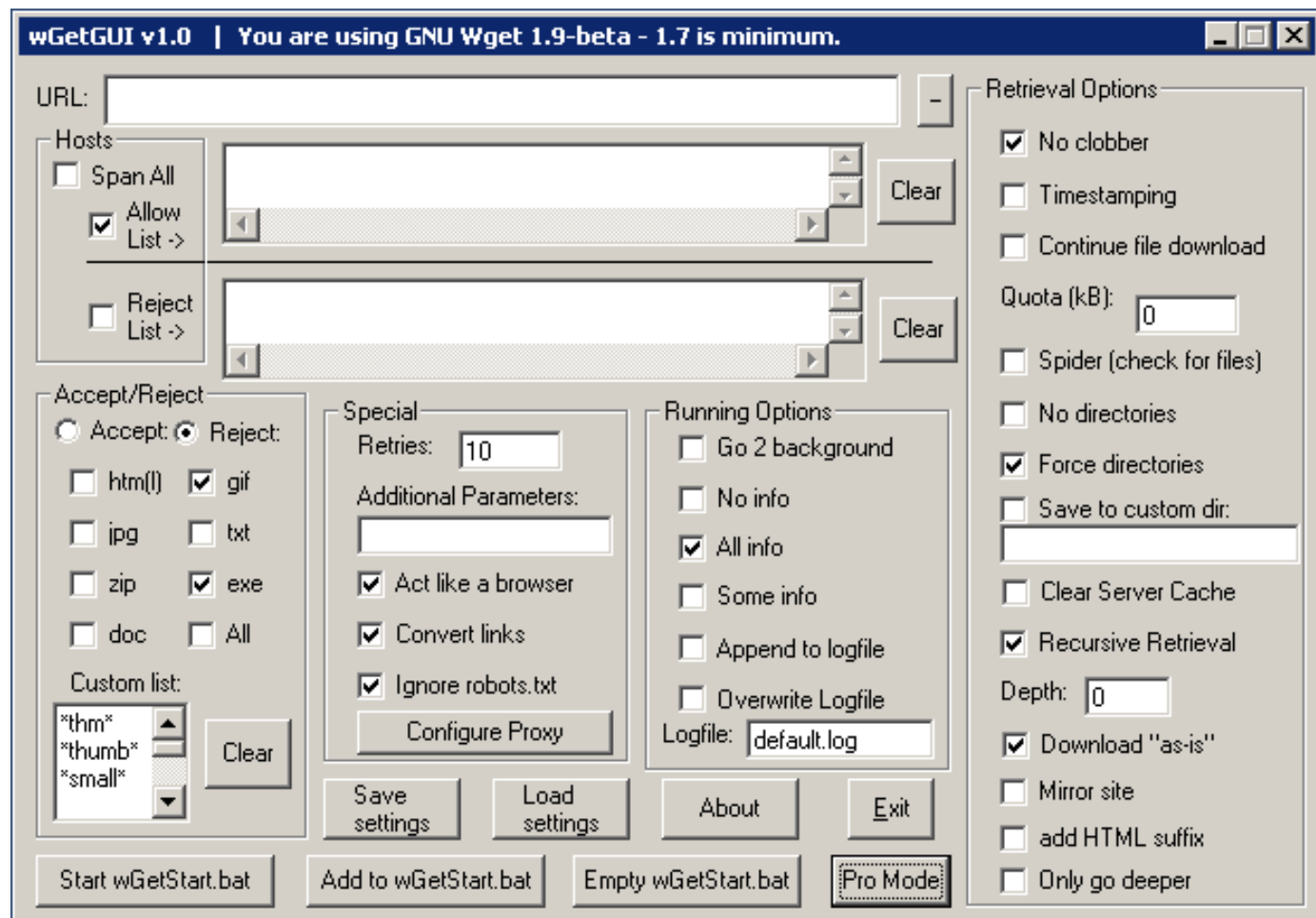


```
ubuntu@GreenWebpage:~/wget_examples$ wget -O docker-install-guide.html https://greenwebpage.com/community/how-to-install-docker-on-ubuntu-24-04/
--2024-05-28 13:50:55-- https://greenwebpage.com/community/how-to-install-docker-on-ubuntu-24-04/
Resolving greenwebpage.com (greenwebpage.com)... 146.70.56.164, 2a04:9dc0:0:a::2
Connecting to greenwebpage.com (greenwebpage.com)|146.70.56.164|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'docker-install-guide.html'

docker-install-guid  [          <=>          ] 145.55K  33.9KB/s  in 4.3s

2024-05-28 13:51:02 (33.9 KB/s) - 'docker-install-guide.html' saved [149048]

ubuntu@GreenWebpage:~/wget_examples$
```





Why?

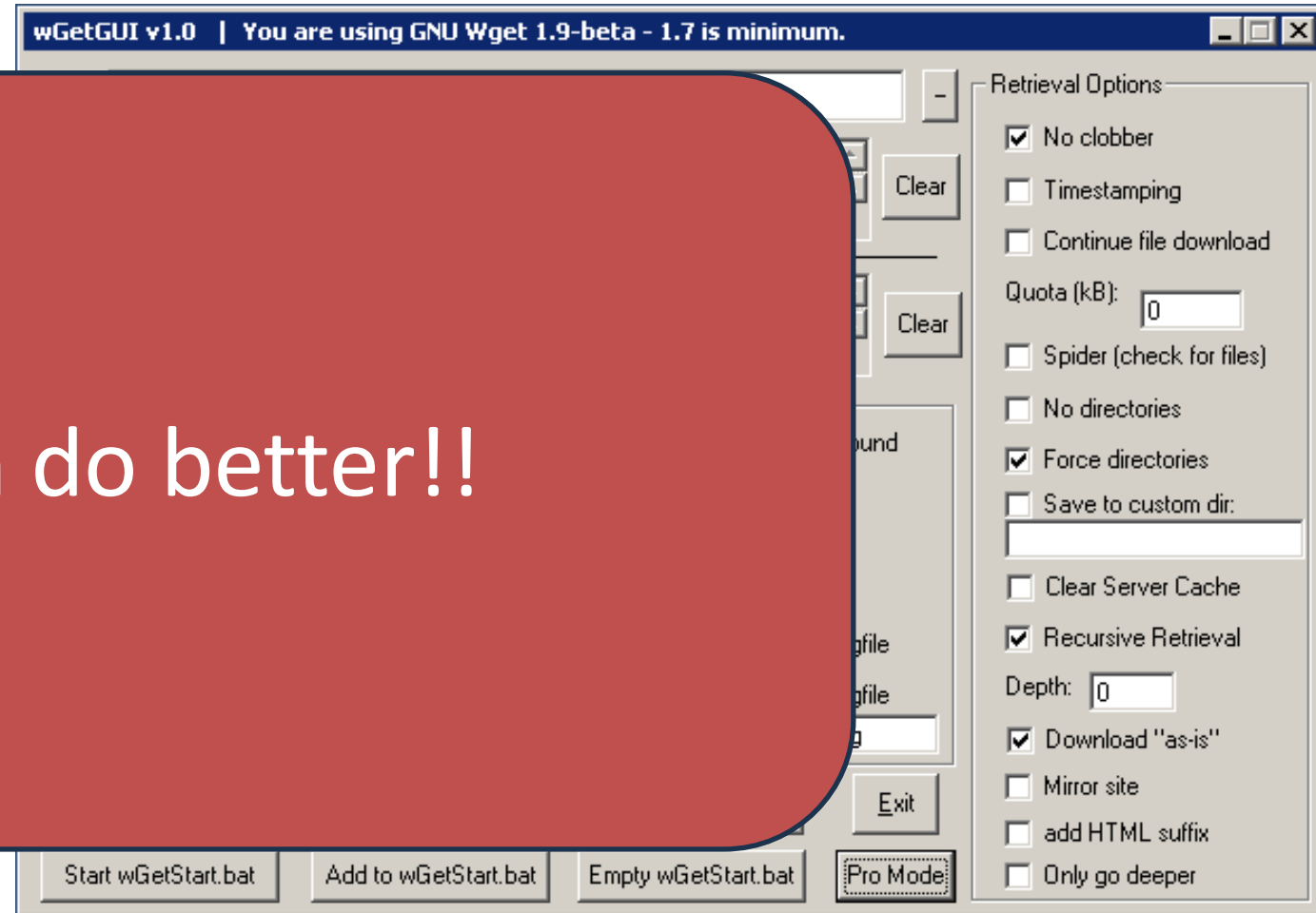
GNU Wget is a free software package for retrieving files using HTTP, HTTPS, FTP and FTPS, the most widely-used protocols. It is a non-interactive **commandline tool** that can be called from scripts and run without X-Window.

```
ubuntu@GreenWebpage:~/wget_example$ wget https://greenwebpage.com/community/how-to-install-docker-on-ubuntu-24-04/
--2024-05-28 13:50:55-- https://greenwebpage.com/community/how-to-install-docker-on-ubuntu-24-04/
Resolving greenwebpage.com (greenwebpage.com): 192.168.1.100.
Connecting to greenwebpage.com (greenwebpage.com): 80.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'docker-install-guide.html'

docker-install-guide.html  [100%]
2024-05-28 13:51:02 (33.9 KB/s)

ubuntu@GreenWebpage:~/wget_example$
```

We can do better!!

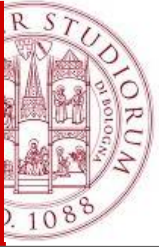




How to Design and Develop Good Interactive Systems?

- *Iterative and human-centered process* Interview must be correct
unless the rest will be affected
- People needs (not “wants”)
- Design principles and guidelines
- Usability goals Usability is a standard
- Prototyping (rapidly and frequently) Low-level (drawing), medium-level
(balsamic) and high-level (figma) prototyping
- Evaluation (various kind)
- Programming

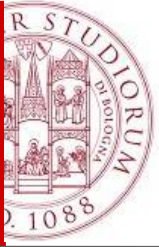
Topics, Organization, and Exam



Course Topics... At a Glance!

1. Introduction to HCI
2. Problem framing and [Find the need of the user](#) needfinding
3. Tasks and their analysis
4. Prototyping at various levels of fidelity
5. Design guidelines, principles, and heuristics
6. Visual design and design patterns
7. Heuristic evaluation
8. Usability testing
9. Advanced interactions [\(Seminars etc etc\)](#)





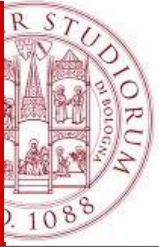
Learning Material

Virtuale

<https://virtuale.unibo.it/course/view.php?id=83400>

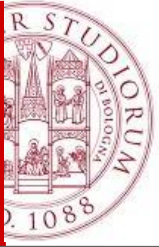
(Automatic Enrollment from Study Plan... or... ?)

- Course materials:
 - Slides, exercises, lab texts
 - Full schedule
 - Templates and deadlines
 - Supplementary material



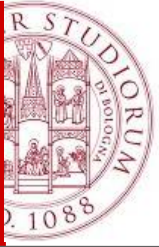
Course Methodology

- Project-based Project done during lab
 - ➔ students learn by doing a project, in teams
- Problem-based
 - ➔ the project work starts from elicited and real users' needs (*needfinding phase*)



Course Methodology

- Projects developed **during** the semester and **step-by-step**
 - Within a given *theme* and mostly in the *labs*
 - Iterating on *prototypes*
- **Feedback** to the teams
 - Feedback is there to help students improve the next step in their projects, in the course, in addition to possibly improve the final grading



Course Organization

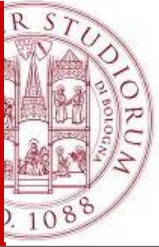
- Classes

- 2 h/week
- Interactive lectures

- Laboratories

- 3 h/week
- Iterating on *prototypes*

	MO	TU	WE	TH	FR
09:00		Lab 4.2			Room 2.5
10:00		Lab 4.2			Room 2.5
11:00		Lab 4.2			
12:00					
13:00					
16:00					
17:30					



Course Organization

■ Classes

- 2 h/week
- Interactive lectures

■ Laboratories

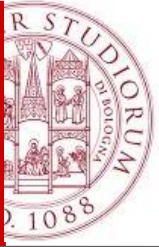
- 3 h/week
- Iterating on *prototypes*

Disclaimer!

It will happen that I need to cancel a few classes. I will always send a notification by Virtuale the week before.

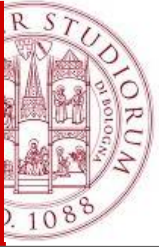
You can still use those hours for the team's work on the project!

	MO	TU	WE	TH	FR
09:00		Lab 4.2			
10:00		Lab 4.2			
11:00		Lab 4.2			Room 2.5
12:00					Room 2.5
13:00					
16:00					
17:30					



Classes

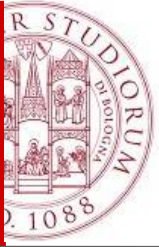
- In-person, in room (2.5)
- Lectures video-recorded and made available on **Panopto**
 - not *streamed live*
 - not in-class exercises and labs



Laboratories

- **Collaborative** and interactive places, to work and share feedback
 - In-person attendance is *fundamental*!
 - If you are a **working student**, send me an email
- **Two main activities** within labs:
 1. **Assignments** - Teams work on steps of the project with the guidance of the teacher
 2. **Checks** - Teams present their assignment work to the teacher and receive feedback

IMPORTANT: Assignments and checks are **informal**! They are just a way to help you in the development of the project, step by step! They are NOT mandatory and don't impact on the final vote!!! These activities will be done during lab hours. Deadlines are not strict (is just for us to have a process step by step)

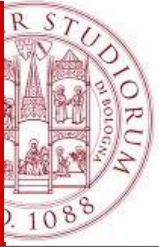


Projects (on laboratories)

Select one theme between the three

- Themes (vertical themes):
 1. Health and Wellbeing
 2. Environmental sustainability
 3. Smart Cities & Urban Life

Choose a broad area to start with: the idea didn't have to be so specific at the start (so the specific smaller area of the broad one)



Theme 1 – Health and Wellbeing

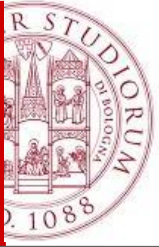
No one: Are you okay?
Me: Yeah. I'm totally fine.
My Phone:

Digital Wellbeing

Check your weekly report

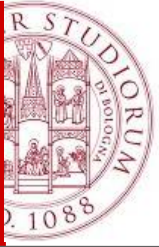
You used your phone 42 h 6 m more than last week.

Description: Health and wellbeing are fundamental aspects of our daily existence. They encompass physical, mental, and emotional dimensions that can be significantly influenced by our interactions with technology. Within this theme, you will explore innovative ways to create applications and interfaces that **empower individuals to lead healthier lives.**



Theme 1 – Health and Wellbeing

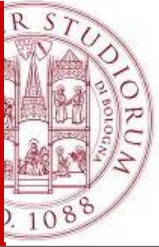
- Examples:
 - Digital wellbeing and healthy technology use
 - Mental wellbeing and stress management
 - Physical activity, sport and fitness
 - Sleep and healthy routines
 - Personal health tracking and self-monitoring
 - Healthy Nutritions
 -



Theme 2 – Environmental sustainability

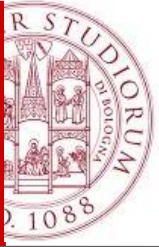


Description: Environmental sustainability refers to using resources in ways that meet present needs without compromising the ability of future generations to meet theirs. It involves protecting ecosystems, reducing pollution, protecting biodiversity, conserving energy and water and shifting toward practices that minimize harm to the planet. Within this theme, we will explore innovative ways to create systems and technology for a more sustainable future, while increasing people awareness!



Theme 2 – Environmental sustainability

- A digital twin application for water monitoring
- A IoT infrastructure for environmental monitoring
- A camera-bases system for fauna detections
- A DIY 3D-printed bees house app for biodiversity monitoring and awareness
- An AI-powered infographic for environmental awareness
- A mobile app for citizen science
-
- ...



Theme 3 – Smart Cities & Urban Life



Description: Cities are complex environments where people, technology, infrastructure, and public services constantly interact. Digital technologies can play an important role in improving how we move, communicate, access services, and experience public spaces. Within this theme, you will explore innovative ways to create applications and interfaces that make urban life more accessible, connected, sustainable, and people-centered.



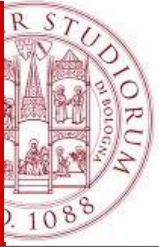
Theme 3– Smart cities & Urban life

- Accessible wayfinding for people with disabilities
- Sustainable smart mobility
- Personalized wayfinding for runners
- Finding accessible and inclusive public spaces
- Connecting citizens with local services
- Supporting citizen reporting of urban issues
- Discovering less crowded routes and places
- ...
- ...

Teams

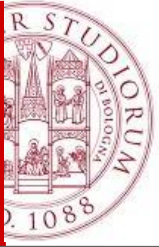
- 3-4 students (*preferably 4*)
 - NB: working students should contact me
- It is students' responsibility to form teams
- Teams cannot be changed during the semester
- Each team will work on their own GitHub repository
 - Each team will create a private repository to share with me





About The Exam

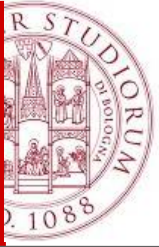
1. **Project development** (up to 20 points)
 - In teams
 - Final report – process, execution, and outcomes of *four group assignments*
 - Prototypes “source”
 2. **Heuristic evaluation** (up to 6 points)
 - Individual
 - Report – outcome and execution of *one individual assignment*
 3. **Oral discussion** on the project (up to 4 points)
 - As a group, mandatory
- The realized project will be valid until the end of the **academic year**
 - Additional points (max 2) can be assigned for the effort during the course, the project quality and creativity



About The Exam

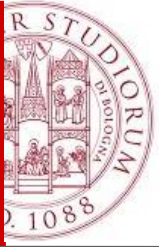
I will provide a *template*
for both reports!

1. **Project development** (up to 20 points)
 - In teams
 - Final report – process, execution, and outcomes of *four group assignments*
 - Prototypes “source”
 2. **Heuristic evaluation** (up to 6 points)
 - Individual
 - Report – outcome and execution of *one individual assignment*
 3. **Oral discussion** on the project (up to 4 points)
 - As a group, mandatory
- The realized project will be valid until the end of the **academic year**
 - Additional points (max 2) can be assigned for the effort during the course, the project quality and creativity, and the oral discussion



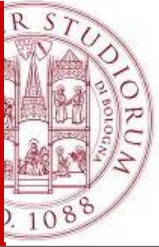
Evaluation Criteria

- Invested effort in the project activity, including the willingness to incorporate the provided feedback
- Originality, complexity, and richness of the work (can fake some part: e.g. faking the interaction with chat bot)
- Methodological and technical correctness of the entire process
- Completeness and communication quality of the assignments' outcomes and report(s)
- Quality of the presentations and oral discussion
- Individual contribution



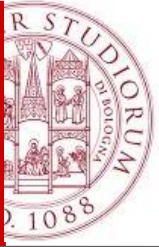
Project Development

- **Goal:**
 - to give hands-on experience with the modern human-centered design process described during the course
- Projects will be built **step-by-step** and mostly carried on *during* labs
- Project's topic proposed by each group
 - Within the chosen theme
 - Based on *needfinding*
- **Group assignments** represent the various process steps
 - **During the labs**
 - Are often followed by *checks with me* (in one of the following labs)
 - **Evaluated only at the exam** through reports and discussion



Assignments and Checks

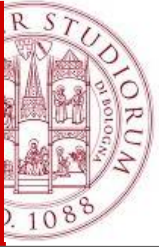
- Assignment 1 [group]
 - *Needfinding* Understanding User needs
- Assignment 2 [group]
 - *Storyboard and Low-fidelity prototype* (Drawing and using Balsamic)
- Assignment 3 [individual]
 - *Heuristic evaluation*
- Assignment 4 [group]
 - *Medium-fidelity prototype*
- Assignment 5 [group]
 - *High-fidelity prototype and usability evaluation*



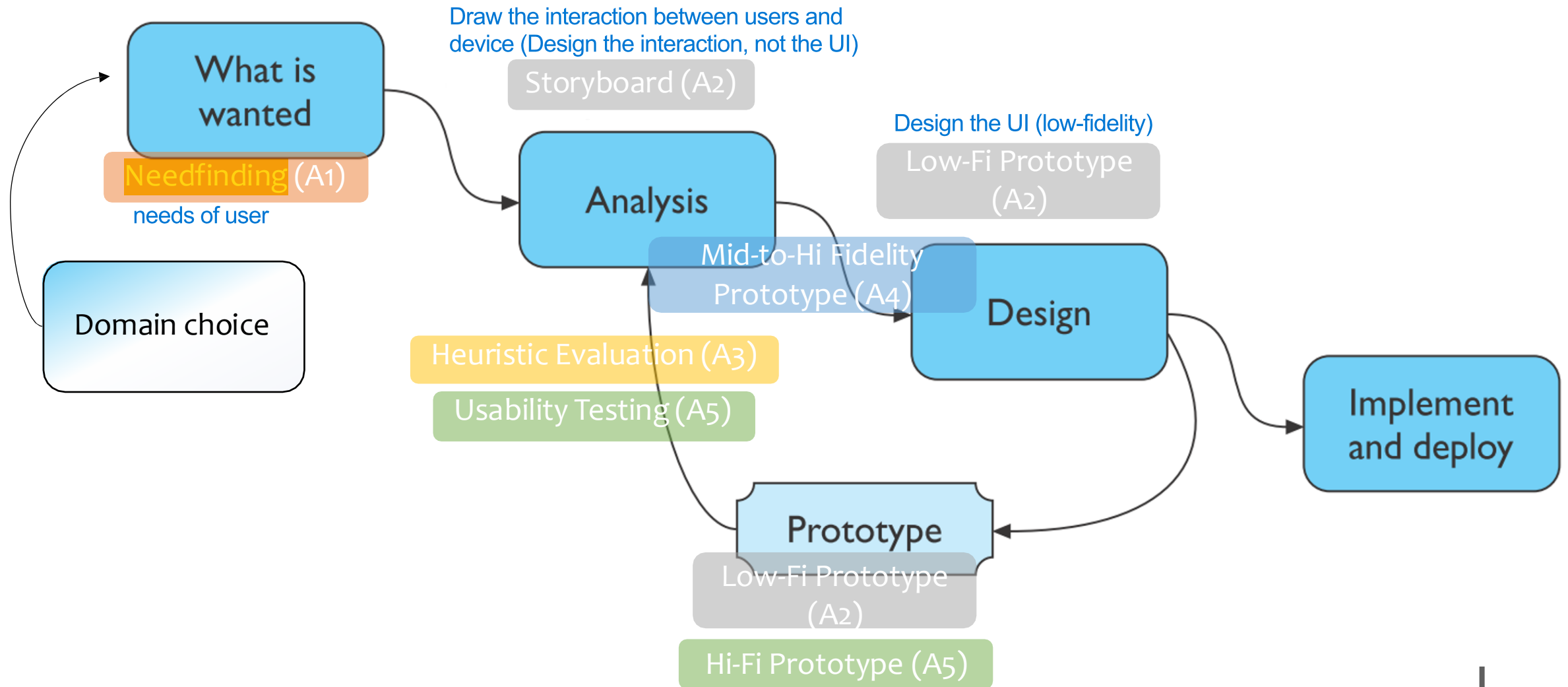
Assignments and Checks

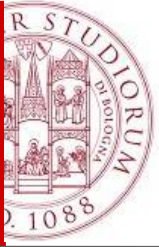
- Assignment 1 [group]
 - *Needfinding*
- Assignment 2 [group]
 - *Storyboard and Low-fidelity prototype*
- Assignment 3 [individual]
 - *Heuristic evaluation*
- Assignment 4 [group]
 - *Medium-fidelity prototype*
- Assignment 5 [group]
 - *High-fidelity prototype and usability evaluation*

Note: not
mandatory during
the class but
evaluate during the
final exam



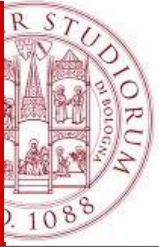
Human-Centered Process vs. Assignments





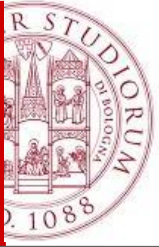
Projects Completion Level (I)

- The realized final prototype *must* be a **high-fidelity interactive prototype**
 - in code, with any technology you like/know
 - not a final “product” (e.g. registration/login can be faked)
 - **simulating** a realistic experience on a specific device
- The application is not required to (fully) implement standard (yet important) features, such as sign-up, sign-in, ...
 - Assume that your user is *already* registered and signed in
- This means that some (difficult or standard) features can be *faked* or *hard coded*



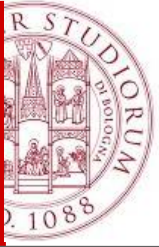
Projects Completion Level (II)

- The prototype will make some assumptions and won't implement standard or very complex features (even if technically important)
 - Which will be *faked*, limited in number, or *hard coded*
 - Your users will be already registered, signed in, and ready to “work” (no sign up, sign in, ... mechanism)
 - Any particular behavior will work in a small number of pre-defined cases
 - ...



Oral Discussion

- **All teammates** present and presenting
- Each group will have 30 mins to:
 1. Give a brief *introduction* to the project (*no slides*)
 2. Do a *demonstration* of the implemented prototype, where students cover the main features and everybody in the team speak
 3. Answer some *questions* about what students showed and/or about the submitted report(s)
- **Beware:** the demonstration is typically the most critical part
 - it needs to be carefully prepared, and not rigged up at the moment
- I will have already read the report(s) and had a look at the final prototype code, so there is no need to cover those



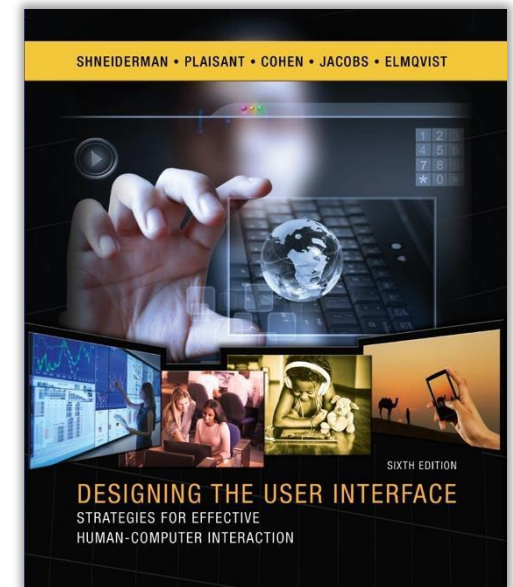
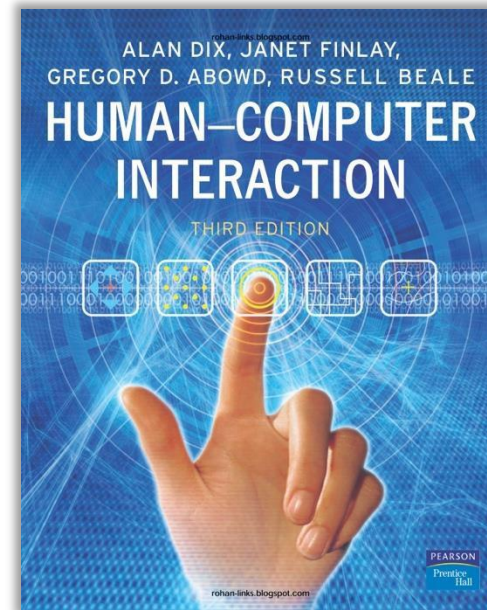
Introducing... the Assignment Zero

- Submit group composition using a wiki page on Virtuale
 - Group name
 - 3 or 4 persons (max), for each:
 - Surname, Name, e-mail

(tentative) Deadline:
September 28, 2026

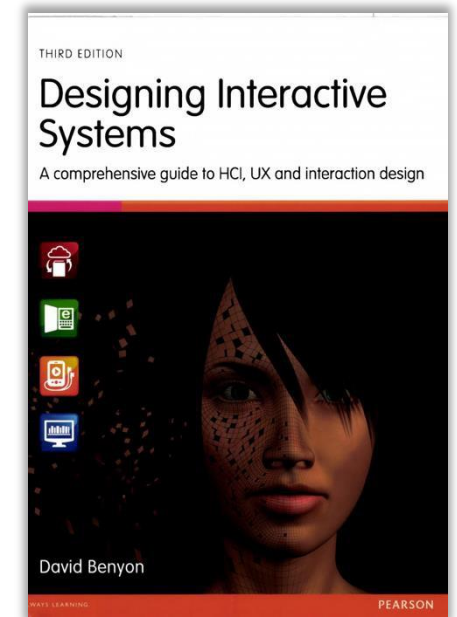
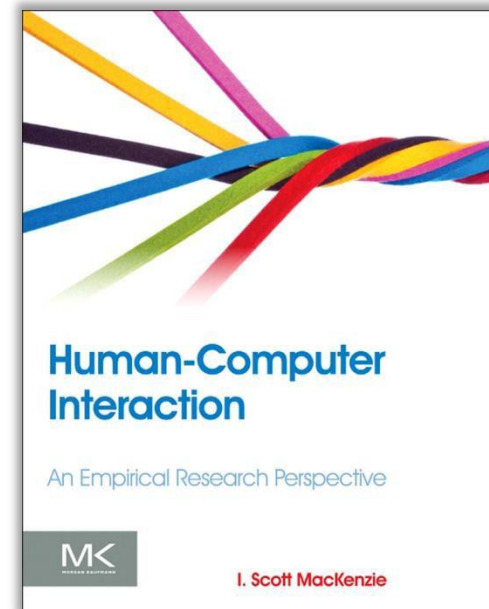
Course Books

- Alan Dix, Janet Finlay, Gregory D. Abowd, Russel Beale, "Human-Computer Interaction", 3rd edition, Prentice Hall, 2004, ISBN 0-13-046109-1
- Shneiderman, Plaisant, Cohen, Jacobs, Elmqvist, "Designing the User Interface: Strategies for Effective Human-Computer Interaction", 6th edition, Pearson, 2016, ISBN 013438038X / 9780134380384



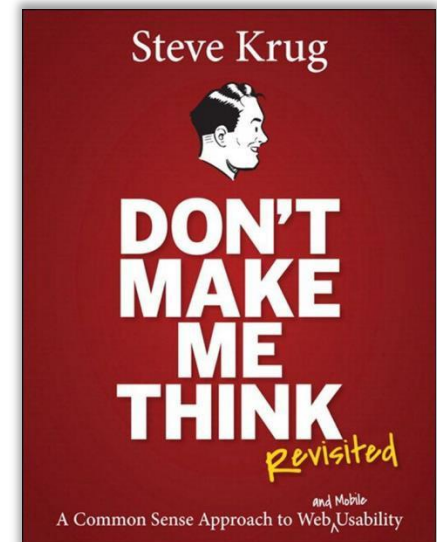
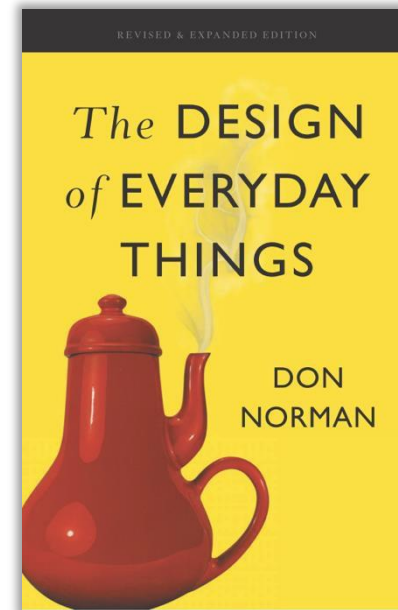
Others Suggested Books

- I. Scott MacKenzie, "Human-Computer Interaction: An Empirical Research Perspective", Morgan Kaufmann, 2013, ISBN 978-0-12-405865-1
- David Benyon, "Designing Interactive Systems", 3rd edition, Pearson, 2014, ISBN 978-1447920113



Others Suggested Books

- Don Norman, "The Design of Everyday Things: Revised and Expanded Edition", Hachette UK, 2013, ISBN 0465072992/9780465072996
- S. Krug, "Don't Make Me Think: A Common Sense Approach to Web and Mobile Usability - revisited", Pearson Education, 2014, ISBN 0321648781/9780321648785



Questions?



These slides are a modified version of the ones of prof. Luigi De Russis, PoliTo.

